

From Wrist to Finger: Hand Pose Tracking Using Ring-Watch Wearables

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Abstract

Hand pose tracking is essential for advancing applications in human-computer interaction. Current approaches, such as vision-based systems and wearable devices, face limitations in portability, usability, and practicality. This paper proposes a novel multimodal hand pose tracking framework that integrates data from an IMU-equipped ring and EMG sensors embedded in a wrist-worn device. By leveraging the complementary strengths of motion dynamics and muscle activity, our deep learning-based sensor fusion approach achieves precise 3D hand pose reconstruction. We fused multichannel data using a transformer-based model incorporating time encoding and cross-modal attention mechanisms. We also designed weighted loss function designed to optimize spatial, kinematic, and anatomical accuracy. Experimental validation using a custom dataset of 19 gestures performed by 10 participants demonstrates robust performance, with an average MPJPE of 0.750 cm and joint angle differences of 6.815° for cross-user evaluation.

CCS Concepts

• **Human-centered computing** → **Systems and tools for interaction design**.

Keywords

Ring, IMU, EMG, Hand Pose Tracking

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1 Introduction and Background

Accurate hand pose tracking is critical for advancing human-computer interaction (HCI) applications, including virtual reality (VR)[28, 29], augmented reality (AR)[7, 22, 33], prosthetics control, and sign language interpretation[14, 31]. The human hand, with its intricate

structure and dynamic range of motion, necessitates precise sensing and modeling techniques to effectively capture its gestures and poses. Among the existing solutions, hand tracking can generally be categorized into vision-based and wearable-device-based approaches.

Vision-based solutions, such as depth camera systems [3, 12] and RGB camera systems [8, 9], while providing high spatial resolution, require external cameras or sensors, which often limit their usability in mobile or privacy-sensitive scenarios. In contrast, wearable-device-based solutions offer a more portable and user-centric alternative, leveraging sensors such as inertial measurement units (IMUs) and electromyography (EMG) for gesture recognition. Traditional wearable solutions include gloves[5, 11, 19], wristbands[4, 20] or armbands[13, 34], and rings[27, 35]. However, these systems typically rely on additional specialized devices that may constrain their practicality in everyday contexts. For example, glove-based systems, such as [21], were design as stretchable but still less suited for daily use. Wristbands or armbands, such as MYO armband [17], often require multi-channel EMG sensors, increasing wearing discomfort and system complexity. Similarly, ring-based solutions, such as OmniRing [37], often demand multiple rings equipped with IMUs or rely on acoustic sensors like microphones and speakers[32], which are less equipped in commercial rings.

Given these limitations, there is a pressing need to explore hand-tracking approaches that utilize existing, widely accepted device forms, enabling more practical and accessible solutions for end users. Addressing this challenge, we propose a novel system in Figure 1(b) that combines a single IMU-equipped ring with EMG sensors embedded in a smartwatch to achieve robust hand pose tracking. Using biomechanical insights and the complementary strengths of IMU and EMG data, IMUs capturing motion dynamics and EMG revealing muscle activity, our approach offers a practical and effective solution for tracking the pose of the hand in the hand.

This paper presents a novel method for reconstructing hand gestures by leveraging the complementary strengths of IMU and EMG sensors. Our approach builds on recent advancements in machine learning for sensor fusion, enabling robust mapping of multi-modal signals to high-dimensional hand pose representations. By addressing key challenges in data integration and pose accuracy, our work contributes to the growing body of research in wearable gesture recognition and reconstruction systems.

The primary contributions of this paper are as follows:

- (1) **Wearable Design:** We introduce a novel ring-watch based multi-channel sensing design equipped with IMU and EMG

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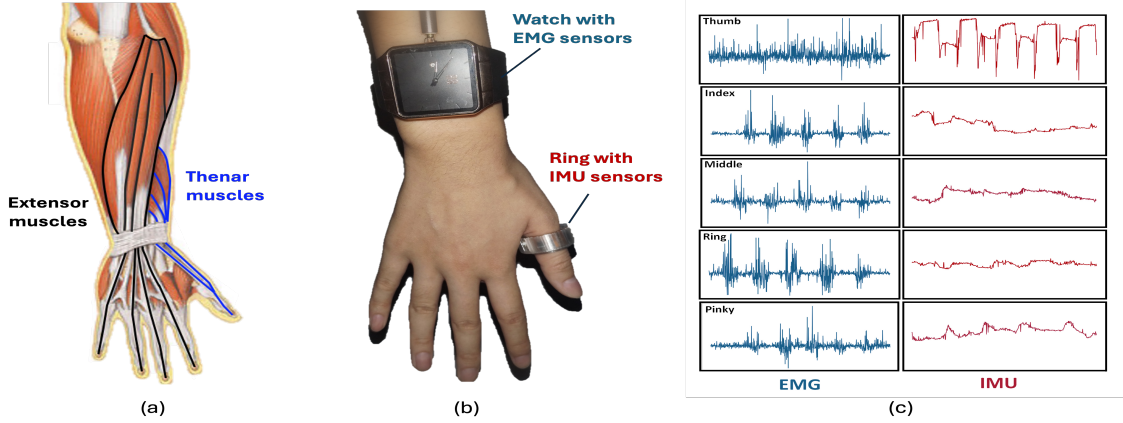


Figure 1: Finger-Wrist Biomechanical Integration. (a) the anatomy of the hand illustrates the distinct muscle groups responsible for movements of thumb and other four fingers. (b) a combination of a watch equipped with EMG sensors and a ring with IMU sensors. (c) when the thumb is moved, the IMU data shows significant changes while the EMG data remains relatively stable. In contrast, when the other fingers are moved, the EMG data displays more pronounced activity, while the IMU data shows only slight, less stable changes.

sensors. This compact and ergonomic design ensures wearability and comfort while accurately capturing finger and hand motion data.

- (2) **Multi-Sensors based Hand Pose Tracking:** Our model integrates IMU signals from the ring and EMG signals from a complementary wrist-worn device to fully capture the dynamics of hand gestures. By leveraging the complementary strengths of kinematic information and muscle activation signals, our deep learning-based fusion architecture enables precise and robust estimation of 3D hand poses across diverse gestures and users.

2 Hand Pose Tracking System

2.1 Biomechanical Sensing Integration of Finger and Wrist

Hand anatomy reveals a distinct biomechanical separation between the muscle groups controlling the thumb and those responsible for the other four fingers. The thumb is primarily governed by the thenar muscles, which include the abductor pollicis brevis, flexor pollicis brevis, and opponens pollicis. These muscles are anatomically and functionally distinct from the extensor muscles located on the dorsal side of the wrist, such as the extensor digitorum communis, extensor indicis, and extensor digiti minimi, which control the extension of the index, middle, ring, and little fingers.

Leveraging this separation, our system integrates data from two wearable sensors: a single-channel electromyography (EMG) sensor embedded in the dorsal side of a smartwatch and an inertial measurement unit (IMU) in a compact ring worn on the thumb. This design capitalizes on the complementary sensing capabilities of EMG and IMU: the EMG sensor captures muscle activation signals from the independent muscle group near the wrist [16], while the IMU tracks the precise kinematic motion of the thumb.

As shown in Figure 1, the integration of these two sensors is particularly effective because it aligns with the natural biomechanical independence of the thumb and the remaining fingers. By focusing on the unique muscle activation patterns and motion characteristics

of the thumb, the combined sensing approach ensures accurate capture of both spatial dynamics and muscle intent. This setup enables robust hand pose tracking, even during complex or high-speed gestures, with minimal hardware requirements.

Unlike prior ring-based approaches that rely on multiple rings worn on different fingers [15, 27, 35, 37] to achieve accurate hand tracking, our method simplifies hardware complexity by requiring only a single ring on the thumb. Additionally, while existing EMG-based solutions often utilize multi-channel EMG [4, 13, 17, 18] arrays for improved signal resolution, such setups introduce integration challenges when embedding them into a smartwatch form factor. In contrast, our approach requires only a single EMG channel, making it highly suitable for seamless smartwatch integration while maintaining effective gesture tracking.

2.2 Wearables Design

2.2.1 Customized Ring Prototype. As shown in Fig.2, the ring prototype consists of a flexible circuitry, an arc-shaped battery, and a 3D-printed case, designed for both compactness and wearability. The prototype is built around the NRF52832 MCU[23], which integrates a 2.4 GHz BLE radio transceiver and an ARM Cortex-M4 32-bit processor in a compact 6mm x 6mm package. The IMU, ICM20948[25], provides 9-axis data and communicates with the MCU via I2C, chosen for its simplicity and efficiency. We use a XC6206P182MR voltage regulator to provide 1.8V for IMU and a AP2112K-3.3TRG1 voltage regulator to provide 3.3V for board. A 2N7002KDW integrating two independent NMOS tubes is used to handle 1.8V and 3.3V connections. To optimize Bluetooth data transmission, the rate is set at 55Hz with a payload of 64 bytes per data package. The companion application has been redesigned to integrate functionalities from the Nordic OTA app[24] and the Adafruit app[1, 2].

Compared to existing smart rings designed for continuous pose tracking (e.g., Ring-a-Pose[32]: 148mW, ssLOTR[36]: 198mW, and PeriSense[30]: 474mW), our ring prototype prioritizes real-time

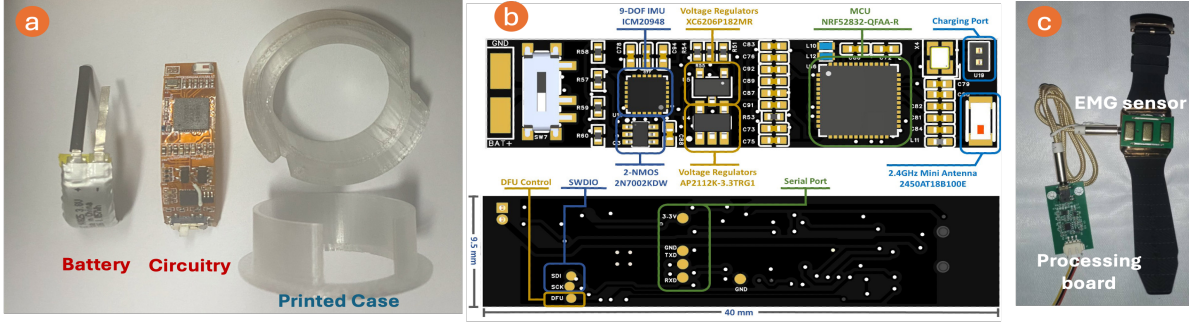


Figure 2: Ring-Watch Wearables. (a) main components of the ring prototype. (b) circuitry of the ring. (c) sensor-side of the watch prototype.

IMU data transmission with a streamlined architecture. By including only essential peripheral circuits, it achieves a more compact design (9.5 mm width, 40 mm length) and lower power consumption (56 mW). In full-featured streaming mode, with real-time data collection and high-rate Bluetooth transmission, the ring operates at a current of 15 mA, enabling continuous operation for over 1 hour on a 15 mAh arc Li-ion battery.

2.2.2 Integrating EMG sensors in a watch. To collect electromyographic data, we used a dry electrode surface electromyography (sEMG) module, which was mounted on a watch to ensure close contact and ease of use as shown in Fig. 2. The sEMG module consists of a dry electrode board and a signal processing board. The dry electrode board includes three gold-plated copper electrodes arranged in parallel, designed to capture differential voltage generated by muscle activity and transmit it via three signal wires to the signal processing board. The signal processing board integrates filtering and amplification circuits, amplifying weak surface electromyographic signals ($\pm 1.5\text{mV}$) by 1000 times[26]. It also effectively suppresses noise, particularly industrial frequency interference, using differential inputs and analog filtering. The signal processing board outputs an analog signal with a range of 0 to 3.0V and a reference voltage of 1.5V. The output intensity reflects the activity level of the monitored muscle. We utilized the ADC pin of an Arduino UNO development board to read this analog signal and transmitted the data via serial communication to a computing device for further processing.

2.3 Multimodal Hand Pose Tracking

We present a multi-sensor fusion framework that combines data from two complementary wearable sensors: IMU sensor embedded in a compact ring for capturing finger motion, and a single-channel EMG sensor attached on the back side of watch to capture hand's muscle activity. By fusing the spatial and kinematic data from the IMU with the muscle intent signals from the EMG, the framework enables accurate and reliable hand pose reconstruction. As depicted in Fig. 3, the framework integrates IMU and EMG sensor data to achieve precise 3D hand pose estimation.

2.3.1 Feature Encoding. The input to our feature encoding module consists of two parts: the IMU data and the EMG data. The IMU data includes 3-axis accelerometer and quaternion information from the thumb and wrist sensors. The EMG data consists of six features derived from the signal: mean, standard deviation, root mean square

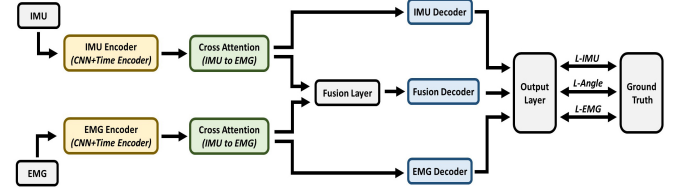


Figure 3: The architecture of the hand pose reconstruction network that integrates IMU and EMG data. (1) The network consists of separate encoders for IMU and EMG signals; (2) cross-attention mechanisms are employed to align IMU data with EMG data. The fusion layer combines both modalities, which are subsequently decoded into separate outputs for IMU and EMG, alongside a fused pose estimate. (3) The output layer generates the final hand pose reconstruction.

(RMS), envelope, mean power spectral density (PSD), and maximum PSD.

To extract meaningful features from the raw sensor data, we designed a feature encoding module consisting of a time encoder and a positional encoder. The time encoder combines a convolutional neural network (CNN) with a Transformer-based encoder. The input data first passes through a 1D convolutional layer to extract local features, which are then fed into the Transformer encoder to model long-range temporal dependencies. The convolutional layer uses three kernels of size 3 and applies a ReLU activation function for non-linear transformation. The Transformer encoder consists of three layers of Transformer encoder blocks, each with four attention heads, effectively capturing temporal dependencies in hand movements and muscle activity. To enhance the understanding of sequential features, we also introduce a positional encoding module with a sinusoidal function to add position information to each time step.

2.3.2 Cross-modal Attention and Feature Fusion. To integrate the multimodal data effectively, a cross-modal attention mechanism dynamically prioritizes the most relevant features from each sensor modality. This mechanism allows the model to align and combine the spatial data from the IMU with the muscle activation signals from the EMG, addressing noise and modality-specific limitations.

The fusion process follows two stages: 1) Relevance Weighting via Cross-Modal Attention: The cross-modal attention is implemented using two dedicated attention blocks, one aligning IMU

features to EMG features, and the other aligning EMG features to IMU features. Each attention block begins by projecting the input features into queries, keys, and values through separate linear layers. These projections are then processed by a multi-head attention mechanism, allowing the model to compute the alignment scores and focus on the most important features in each modality. The output of the attention mechanism undergoes further refinement through an additional linear layer to ensure dimensional consistency; 2) Feature Integration: The cross-modal attention is implemented using two dedicated attention blocks: one aligning IMU features to EMG features, and the other aligning EMG features to IMU features. Each attention block begins by projecting the input features into queries, keys, and values through separate linear layers. These projections are then processed by a multi-head attention mechanism, allowing the model to compute the alignment scores and focus on the most important features in each modality. The output of the attention mechanism undergoes further refinement through an additional linear layer to ensure dimensional consistency.

2.3.3 Hand Pose Reconstruction and Loss Function. The fused features are passed through fully connected layers to predict the final 3D hand pose. To optimize the sensor fusion process and enhance reconstruction accuracy, we design a weighted loss function that integrates contributions from IMU data, EMG signals, and joint angles. Each loss component is tailored to address specific aspects of hand pose reconstruction, ensuring complementary strengths from each data modality.

The IMU loss captures spatial and kinematic information by calculating the difference between the predicted and ground truth hand poses. By leveraging the high temporal resolution of IMU data, this loss ensures precise tracking of hand movements, especially for dynamic motions. The EMG loss evaluates the alignment between predicted hand poses and muscle activation patterns detected by the EMG sensor. This component incorporates intent-driven information, enabling the model to infer subtle finger and hand movements that may not be directly observable from kinematics alone. The angle loss measures discrepancies in 3D joint angles, ensuring that the reconstructed hand poses adhere to anatomical constraints. This loss term not only improves the physical plausibility of predicted poses but also enhances robustness under noisy or incomplete data conditions.

By combining these three loss components, the model effectively balances spatial precision, muscle intent, and anatomical consistency, resulting in a robust and accurate hand pose reconstruction framework.

The total loss function is expressed as:

$$L_{\text{total}} = \lambda_{\text{IMU}}L_{\text{IMU}} + \lambda_{\text{EMG}}L_{\text{EMG}} + \lambda_{\text{Angle}}L_{\text{Angle}}$$

where λ_{IMU} , λ_{EMG} , and λ_{Angle} are weighting factors used to balance the influence of each loss component.

We added an output layer at the end of the model to decode the joint location information. Specifically, the output layer's dimension is set to $\text{num_joints} * 3$, where $\text{num_joints}=21$, corresponding to the 3D coordinates of each joint. This layer generates the position coordinates of each joint from the encoded features, which are ultimately used for predicting the 3D hand pose.

3 Evaluation

3.1 Experiments

Data Collection: Data was collected from 10 participants performing 19 distinct hand gestures (as shown in the Appendix). Each participant performed each gesture continuously five times during the data collection process, with a total duration of 30 seconds. Therefore, the average duration for each gesture was 6 seconds. These consecutive repetitions provided multiple sample data points for each gesture. Each participant's dataset included synchronized recordings from an IMU ring, a wrist-worn EMG sensor, and ground-truth 3D hand poses obtained using a motion capture system.

Model Implementation: The model was trained in multiple stages to achieve optimal performance. Initially, the model was trained end-to-end to establish a robust baseline, followed by fine-tuning specific components, such as the pose decoding layer and the EMG encoder, to enhance precision. Hyperparameters were carefully tuned to balance the contributions of the IMU, EMG, and angular loss components, with optimal values of $\lambda_{\text{IMU}} = 0.5$, $\lambda_{\text{EMG}} = 0.5$, and $\lambda_{\text{angle}} = 1.0$. This ensured a harmonious trade-off between modalities and effective minimization of angular discrepancies. Additionally, learning rates of 0.001 for training and 0.0001 for fine-tuning, paired with a patience-based learning rate scheduler, facilitated stable convergence while avoiding overfitting. These adjustments collectively enabled the model to achieve precise and robust 3D hand pose estimation.

Evaluation Metrics: The proposed framework's performance was assessed using established metrics, including Mean Per Joint Position Error (MPJPE) and joint angle differences. These metrics comprehensively evaluate the accuracy of 3D pose reconstruction, as well as the anatomical consistency of joint movements.

3.2 Experimental Results

3.2.1 Tracking Performance Among Different Fingers. Fig. 4(a) illustrates the tracking accuracy for each finger, with the thumb achieving the lowest MPJPE (0.625 cm) and joint angle difference (5.9°), while the little finger showed the highest errors (MPJPE: 0.935 cm, angle difference: 7.8°). The results highlight the impact of sensor proximity, with fingers closer to the IMU (e.g., thumb, index) exhibiting better performance. Anatomical variations and reduced motion range of smaller fingers, such as the little finger, contributed to increased errors. These findings underscore the need for sensor placement and algorithmic refinement to ensure consistent performance.

3.2.2 Leave-One-User-Out Cross-Validation. We conducted leave-one-user-out cross-validation to evaluate the generalizability of our framework. Figure 4(b) illustrates the results for each participant, showing both the MPJPE and joint angle differences as bar charts. The blue and orange dashed lines represent the average performance across all participants for each metric. The results demonstrate the model's robustness and accuracy across participants. MPJPE values ranged tightly from 0.520 cm to 0.934 cm, while joint angle differences remained consistently below 10°, with an average of 6.815°. This consistency underscores the framework's ability to generalize across diverse hand anatomies and gestures,

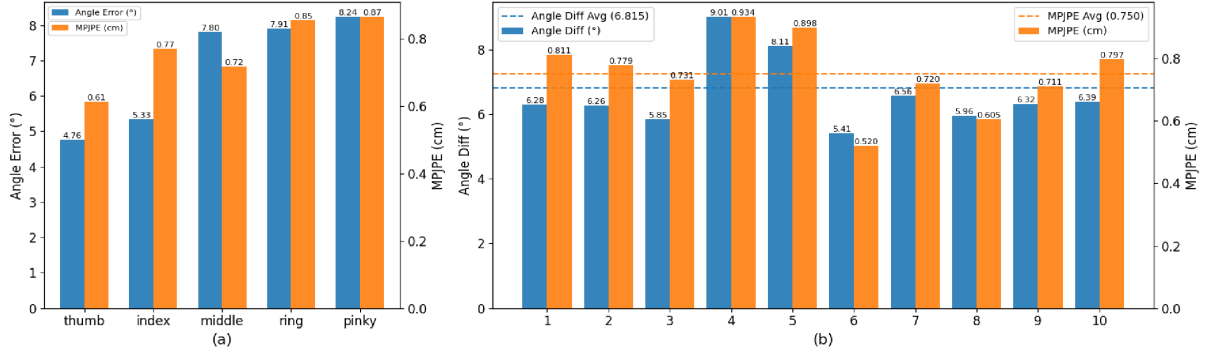


Figure 4: (a) Tracking Errors among different fingers (b) Leave-One-User-Out Cross-Validation Results. The bar charts show the MPJPE (cm) and joint angle differences (°) for each participant, with the red dashed lines indicating the average values.

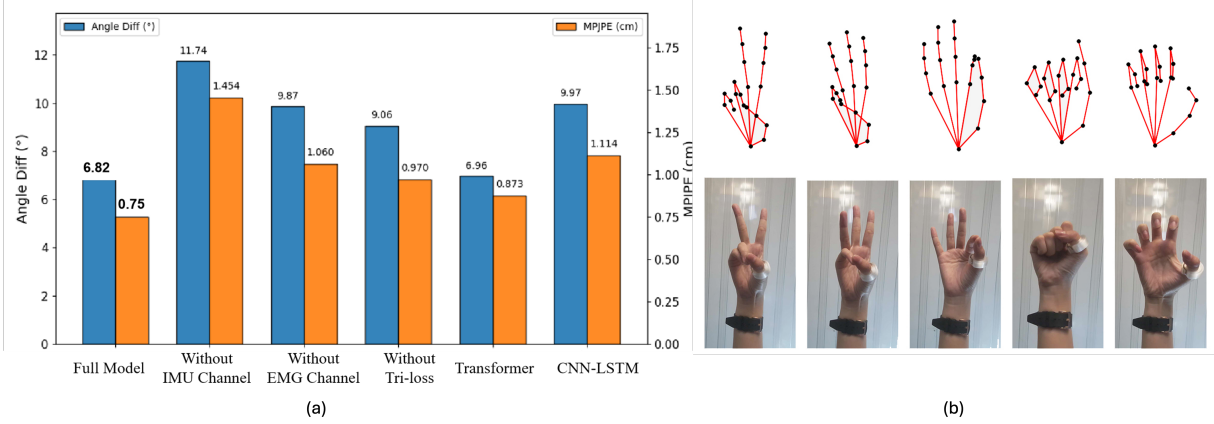


Figure 5: (a) Ablation study results. (b) Qualitative demonstration of five different gestures.

likely due to the complementary integration of IMU and EMG data. These findings highlight the framework’s potential for reliable cross-user hand pose reconstruction in HCI applications.

3.2.3 Ablation Study. To evaluate the contributions of individual components, we performed ablation studies by systematically removing key modules, including the input of IMU, the input of EMG, the loss optimization components, and the cross-modal attention mechanism. Figure 5(a) illustrates the results of these experiments.

The ablation study results demonstrate the importance of each component in the proposed framework. The full model achieves the best performance with an angle difference of 6.815° and MPJPE of 0.750 cm. Removing the IMU module causes the most significant degradation, with an angle difference increasing to 11.744° and MPJPE to 1.454 cm, highlighting the critical role of IMU data. The EMG module also proves essential, as its removal raises the angle difference to 9.868° and MPJPE to 1.060 cm. Cross-attention contributes to effective multimodal fusion, as removing it increases MPJPE to 0.873 cm. These findings validate the necessity of all components in achieving optimal performance.

4 Discussion

4.1 Applications

Figure 5(b) illustrates examples of reconstructed 3D hand poses captured using the ring-watch wearable system. This work establishes

a connection between everyday wearable devices and high-fidelity gesture reconstruction, demonstrating its potential for a wide range of applications.

4.1.1 Dexterous Hand Control. To explore the applicability of the hand pose tracking results, we utilized the uHand UNO platform[10] as a testbed. The uHand UNO is an open-source robotic hand built on the Atmega328 platform, featuring a 6-channel knob controller, a Bluetooth module, six anti-blocking servos, and other high-performance hardware components. We used a Raspberry Pi as the control hub to interface with the uHand UNO, enabling wireless communication and real-time execution of predefined action sets. Each action specifies the completion status and finger bending angles (0° to 180°), allowing precise execution of individual or sequential actions. Algorithms [6] were employed to map the finger positions predicted by our hand pose tracking model to the robotic hand’s corresponding angles. The resulting actions, as shown in Fig.6(a), demonstrate close alignment between the predicted values and the expected human hand posture.

4.1.2 Virtual and Augmented Reality (VR/AR). The proposed hand pose tracking system significantly enhances user interaction in virtual and augmented environments. This advancement supports a wide range of applications, from immersive gaming to professional

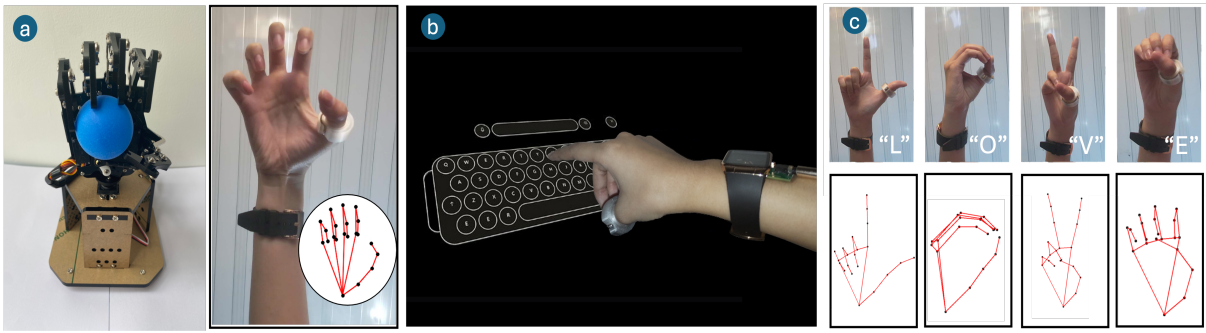


Figure 6: (a) Controlling a dexterous hand based on reconstructed hand poses. (b) Virtual keyboard input for gesture-based interaction. (c) Sign language recognition, demonstrated with the gestures "L", "O", "V", and "E" alongside their corresponding reconstructed hand poses.

AR tools. For instance, the system facilitates accurate virtual keyboard input, allowing users to type or interact with digital interfaces seamlessly in VR settings (Fig. 6(b)).

4.1.3 Assistive Technology. In the realm of assistive technology, the system empowers users by providing accurate and fine-grained hand gesture recognition, improving accessibility and communication tools. For example, its ability to recognize detailed sign language gestures can enable real-time interpretation (Fig. 6(c)).

4.2 Future Work

Future efforts will focus on improving the wearability and versatility of the system while expanding its functional capabilities. Specifically, the development of lightweight and flexible EMG sensors using stretchable polymers. Furthermore, advancing the system's capabilities beyond precise hand pose tracking to include the reconstruction of mechanical data, such as grip force, contact torque, and finger joint dynamics, opens up new possibilities for applications in robotics, ergonomics, and virtual reality. Another promising direction involves integrating haptic feedback mechanisms to enable bidirectional interaction, further enriching the user experience in virtual environments.

5 Conclusion

This study proposes a wearable system for real-time 3D hand pose tracking using an IMU-equipped ring and a wrist-mounted EMG sensor. By leveraging cross-modal attention and a robust loss function, the system achieves robust performance, with an average MPJPE of 0.750 cm and joint angle differences of 6.815° , demonstrating strong spatial and anatomical accuracy across diverse users and gestures. The wearable design prioritizes comfort and practicality, addressing limitations of existing solutions. Its robust results highlight potential applications in VR/AR, human-computer interaction, and assistive technologies. Future efforts will enhance its adaptability and real-world robustness.

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A Appendix

A.1 Gestures Set

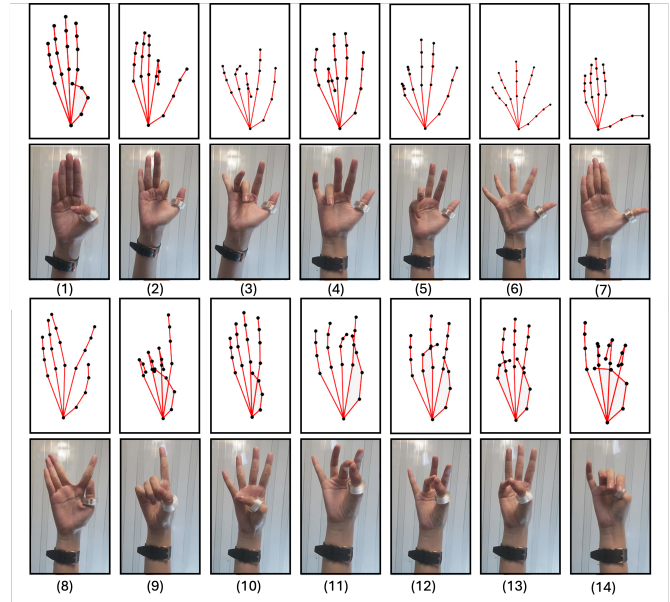


Figure 7: The remaining 14 hand gestures.